

MICRO-DEGREE

Künstliche Intelligenz und Gesellschaft

***Artificial Intelligence
and Society***



Micro-Degree Artificial Intelligence and Society

– Technical Aspects of AI –

Lecture 3.2: Distance & Similarity Measures

Jana Lasser

Research group

Complex Social & Computational Systems (CS)²

IDEa_lab

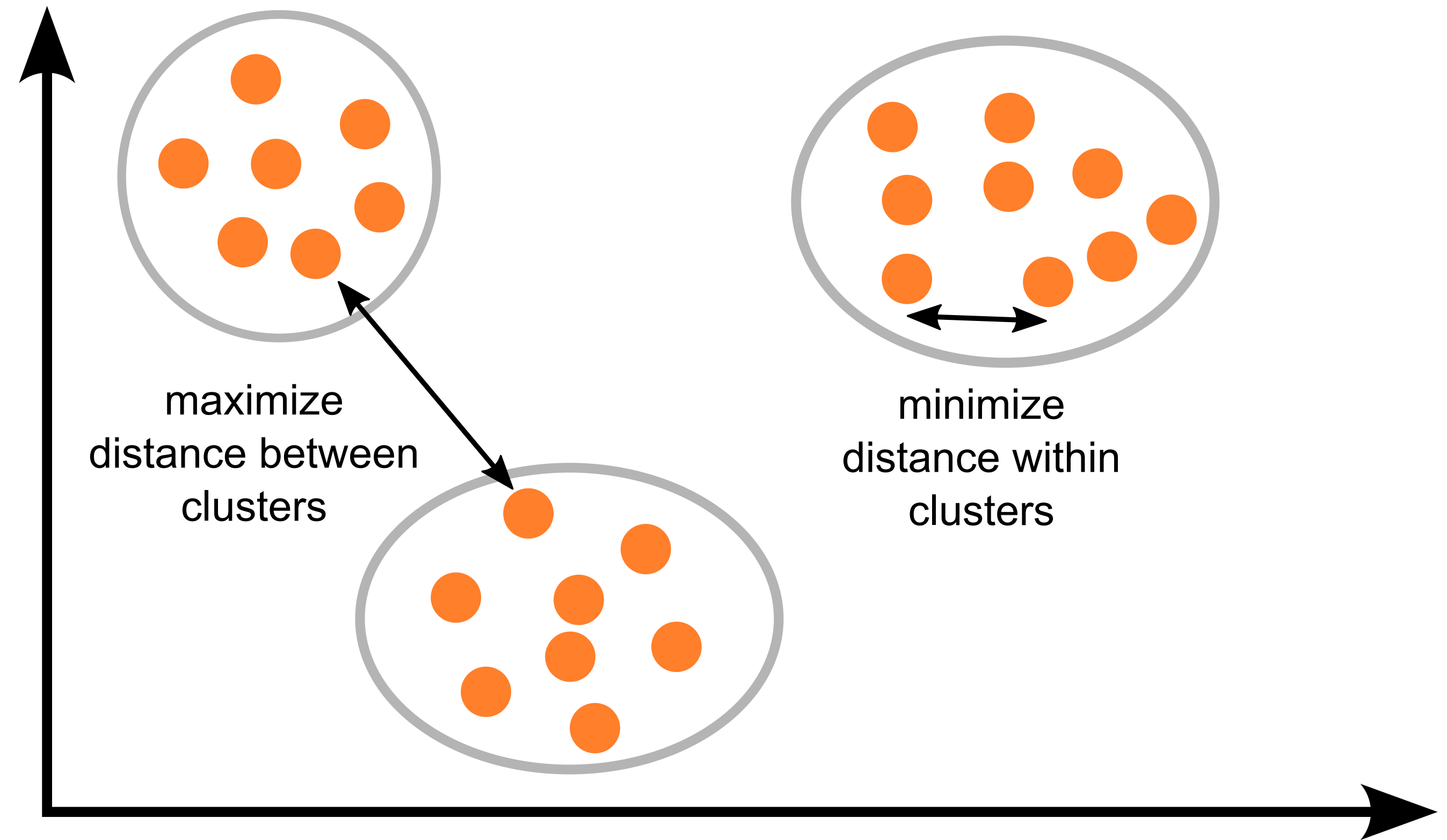
Das Interdisziplinäre Digitale
Labor der Universität Graz



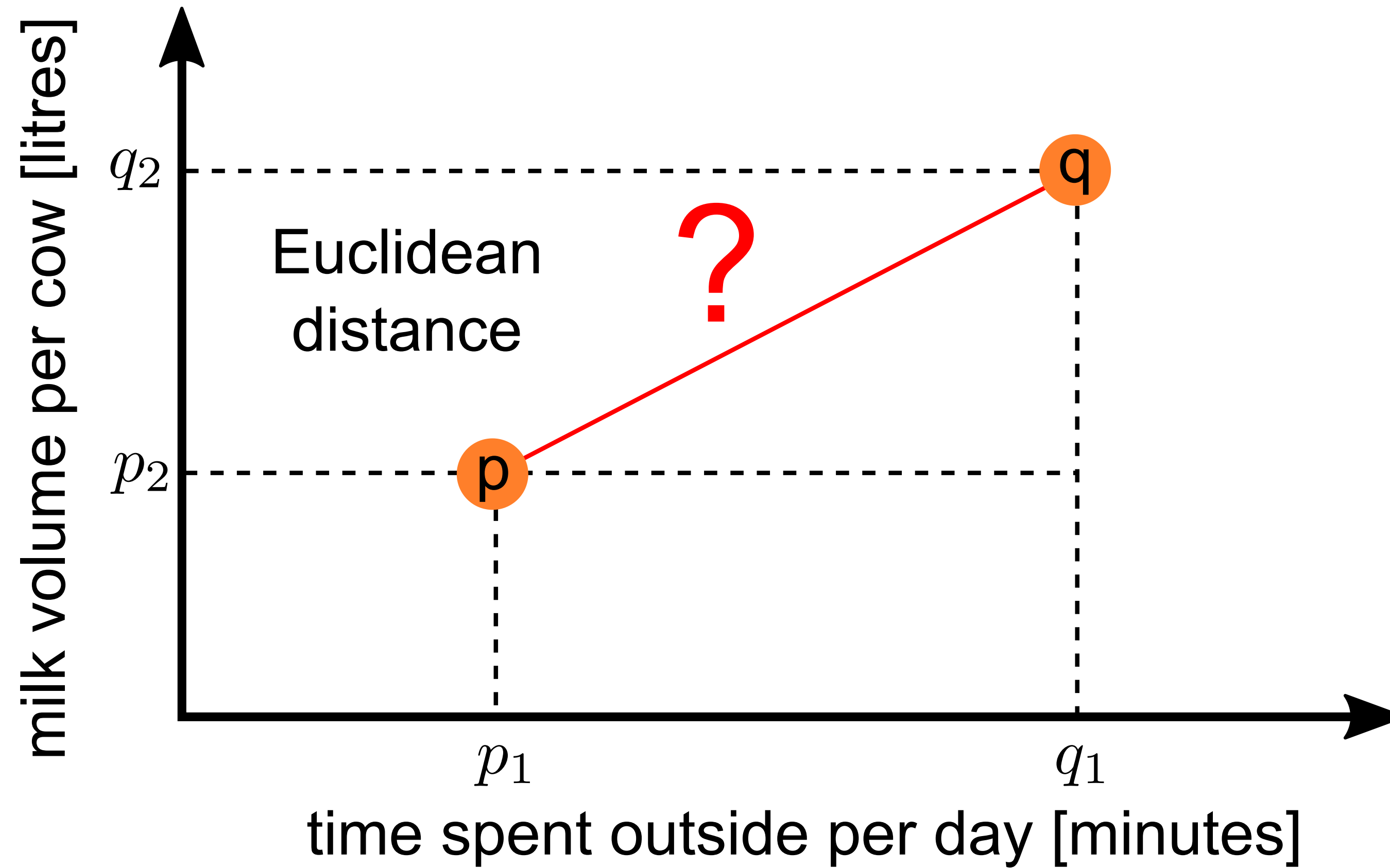
How do we define "distance"?

Euclidean distance is an intuitive *measure* of distance.

The greater the distance, the less **related** two data points are.

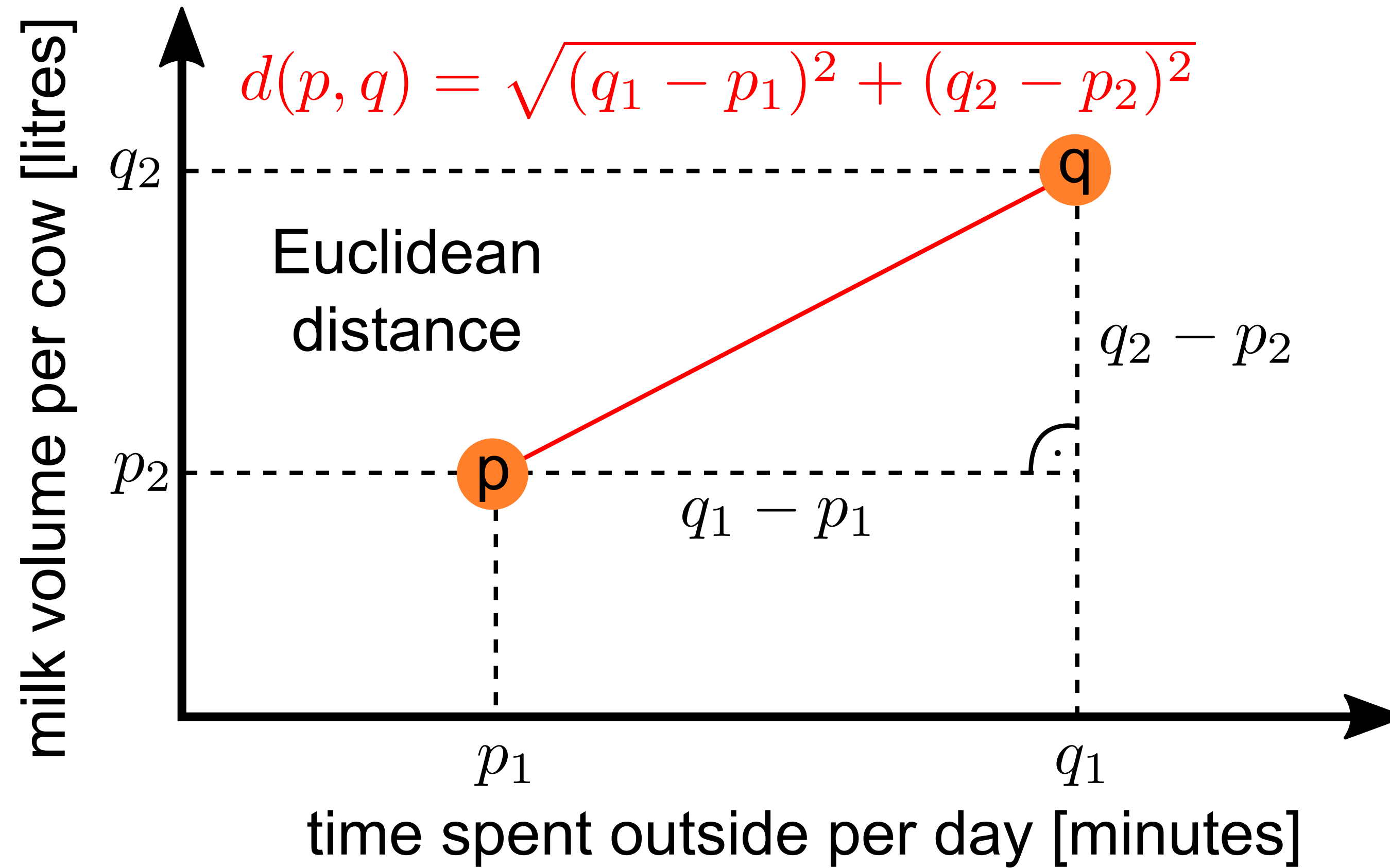


Euclidean Distance in 2 Dimensions



farm	milk	outside
q	112	360
p	55	100

Euclidean Distance in 2 Dimensions



farm	milk	outside
q	112	360
p	55	100

Euclidean Distance in n Dimensions

When observations have more than three attributes, intuitive visualization is difficult.

Example Euclidean distance with 4 dimensions:

farm	milk [L]	outside [min]	altitude [m]	size [# cows]
q	112	360	340	1200
p	55	100	600	350

$$d(p, q) = \sqrt{(112 - 55)^2 + (360 - 100)^2 + (340 - 600)^2 + (1200 - 350)^2}$$

Euclidean Distance in n Dimensions

When observations have more than three attributes, intuitive visualization is difficult.

Example Euclidean distance with 4 dimensions:

farm	milk [L]	outside [min]	altitude [m]	size [# cows]
q	112	360	340	1200
p	55	100	600	350

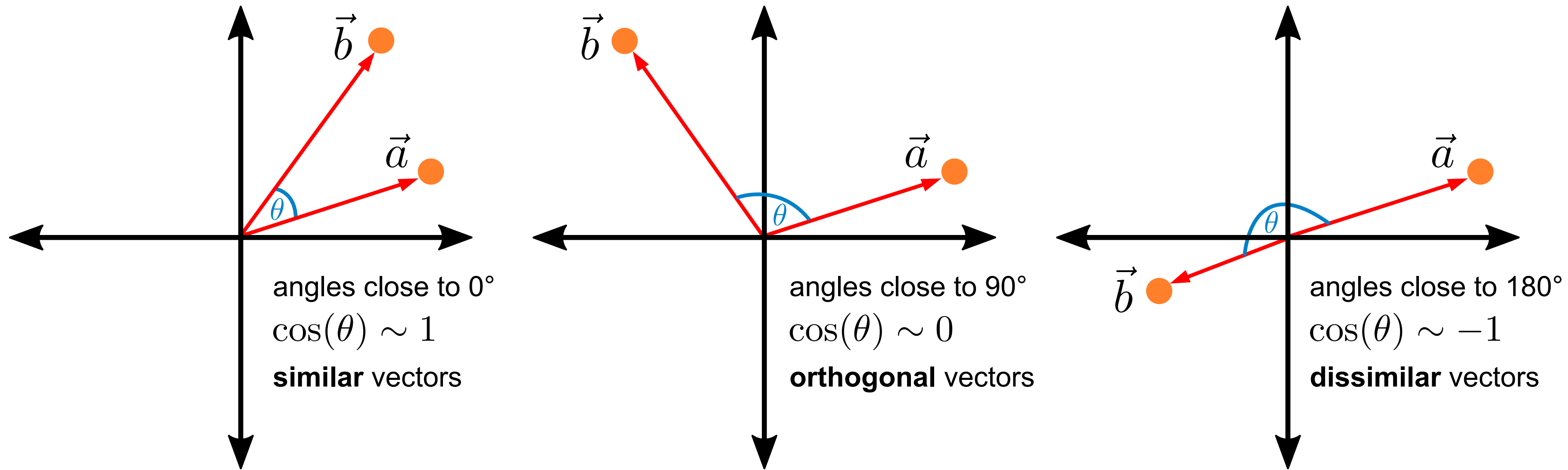
The Euclidean distance between two points $d(p, q)$ can be generalized to any number n of dimensions:

$$d(p, q) = \sqrt{(q_1 - p_1)^2 + (q_2 - p_2)^2 + \dots + (p_n - q_n)^2}$$

$$d(p, q) = \sqrt{\sum_{i=1}^n (q_i - p_i)^2}$$

Cosine Similarity

For observations with a high number of attributes, cosine similarity is suitable. Why? → see "Curse of Dimensionality" later in the lecture! Source: inspired by LearnDatasci, [Cosine Similarity](#).

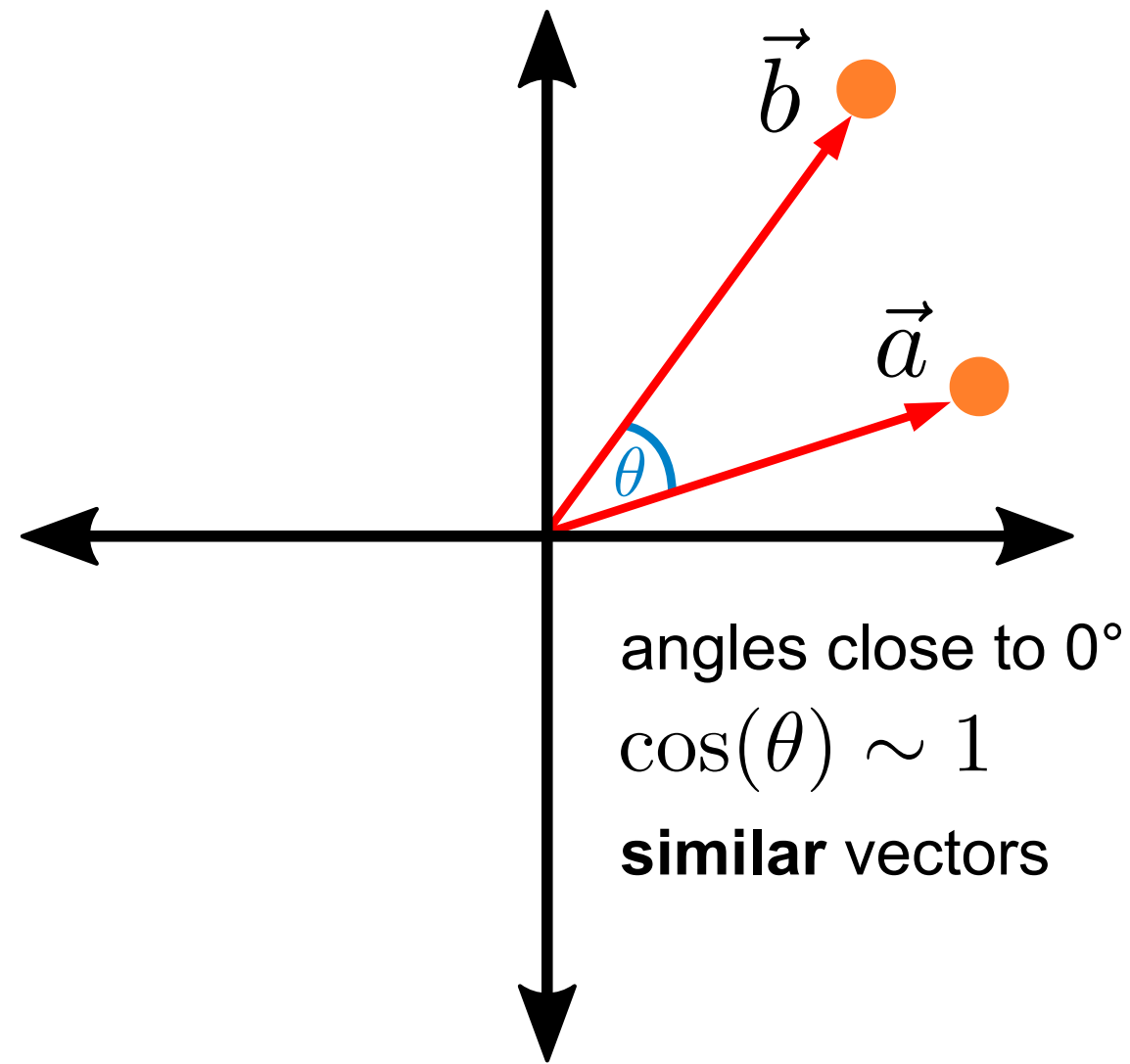


$$S_C(\vec{a}, \vec{b}) := \cos(\theta) = \frac{\vec{a} \cdot \vec{b}}{||\vec{a}|| \ ||\vec{b}||} = \frac{\sum_{i=1}^n a_i b_i}{\sqrt{\sum_{i=1}^n a_i^2} \cdot \sqrt{\sum_{i=1}^n b_i^2}}$$

Cosine similarity

Cosine Similarity

For observations with a high number of attributes, cosine similarity is suitable. Why? → see "Curse of Dimensionality" later in the lecture! Source: inspired by LearnDatasci, [Cosine Similarity](#).



farm	milk x	outside y
a	100	20
b	90	80

$$\vec{a} = \begin{pmatrix} 100 \\ 20 \end{pmatrix} \quad \vec{b} = \begin{pmatrix} 90 \\ 80 \end{pmatrix}$$

$$S_C(\vec{a}, \vec{b}) = \frac{\sum_{i=1}^n a_i b_i}{\sqrt{\sum_{i=1}^n a_i^2} \cdot \sqrt{\sum_{i=1}^n b_i^2}}$$

$$= \frac{100 \cdot 90 + 20 \cdot 80}{\sqrt{100^2 + 20^2} \cdot \sqrt{90^2 + 80^2}} = \frac{10600}{102.98 \cdot 120.42} = 0.86$$

Similarity Measures for Binary Attributes

B \ A	0	1
0	value 0 for A and B	value 1 for A, 0 for B
1	value 0 for A, 1 for B	value 1 for A and B

	farm	hay	milking machine
A		1	1
B		1	0

B \ A	0	1
0	M_{00}	M_{10}
1	M_{01}	M_{11}

Jaccard similarity

$$J = \frac{M_{11}}{M_{01} + M_{10} + M_{11}}$$

Similarity Measures for Binary Attributes

Exercise: calculate J

farm	hay	milking machine
A	1	1
B	1	0

B \ A	0	1
0	M_{00}	M_{10}
1	M_{01}	M_{11}

Jaccard similarity

$$J = \frac{M_{11}}{M_{01} + M_{10} + M_{11}}$$

Attribute has more than two possible values? → One-hot encoding first.

Further Distance & Similarity Measures

Binary attributes

- Simple matching coefficient
- Russell & Rao coefficient

Numbers

- Manhattan Distance

See K. Backhaus et al., [Multivariate Analysemethoden](#), pages 438-453 for details.

Summary

- To apply cluster analysis, we need to establish a measure for the **distance** or **similarity** of observations to each other.
- Distance measures for numbers
 - Euclidean Distance
 - Cosine Similarity
 - Pearson Correlation
- Distance measures for binary attributes
 - Jaccard Similarity
 - Simple Matching Coefficient
 - Russel & Rao Coefficient